ECON 304 – Intermediate Macroeconomics

Problem Set 4

Instructor: Sean Callahan | Fall 2026 | 100 points

Instructions: Show all work clearly. Partial credit may be awarded for correct methodology even if the final answer is incorrect. Groups of no more than three are permitted.

Problem 1: Production Functions (40 pts)

Question 1.1 (10 pts)

Label the following as either constant (CRS), decreasing (DRS) or increasing returns to scale (IRS) with the given production function.

1. $Y_t = L_t^{0.5} K_t$
2. $Y_t = 3L_t^{0.3} K_t^{0.4}$
3. $Y_t = 0.5L_t^{0.6} K_t^{1.2}$
4. $Y_t = 0.4 \log(L_t) + 0.6 \log(K_t)$
5. $Y_t = 0.2L_t + 0.7K_t$

Question 1.2 (15 pts)

Take the first **and** second derivatives with respect to capital of the production functions from Question 1.1. Graph the marginal product of capital (MPK) for each (capture the general shape of the derivative).

Question 1.3 (15 pts)

A core requirement for the standard Solow Growth Model is that the aggregate production function exhibits Constant Returns to Scale (CRS).

Part A (5 pts): Using the production function $Y_t = 0.2L_t + 0.7K_t$, prove that it can be rewritten entirely in *intensive form* (per-worker terms), $y_t = f(k_t)$, where $y_t = \frac{Y_t}{L_t}$ and $k_t = \frac{K_t}{L_t}$.

Part B (10 pts): Now, attempt to rewrite the production function $Y_t = 3L_t^{0.3}K_t^{0.4}$ in intensive form. Explain mathematically why this fails, and briefly discuss what a failure to reach intensive form implies about the role of population size in that economy.

Problem 2: The Solow Model (60 pts)

For Questions 2.1 and 2.2, assume $F(L, K_t) = L^{3/4}K_t^{1/4}$, $s = 0.2$ *and* $\delta = 0.1$.

Question 2.1 (10 pts)

Given what you know about aggregate output, derive the capital accumulation equation (or capital motion equation) by relating the future capital stock to today's investment/saving decision s and the depreciation of capital δ. Express this in terms of labor (i.e., per worker). Assume the government runs a balanced budget.

Question 2.2 (30 pts)

Solve for the steady-state level of capital per worker k^* and output per worker y^*. What is the output-capital ratio and is it constant? What is the growth rate of output and the growth rate of capital? How, if at all, do they differ? Finally, what does the level of economic growth mean for workers in the long run?

Question 2.3 (30 pts)

Suppose the economy is characterized by a CRS production function of the form: $F(A_tL_t, K_t) = (A_tL_t)^{\alpha}K_t^{\beta}$. The coefficient A_t represents total factor productivity. If the growth rate of A_t is zero ($a = 0$), then long-run economic growth converges to the growth rates derived in the previous question. Suppose this mythical country experiments with institutional reforms that promote innovation and international competition such that technological progress begins to emerge and accelerate ($a > 0$).

This technical progress is qualitatively *labor-augmenting*. What is the growth rate of output-per-effective worker $\tilde{y}^*$ and output-per-worker y^*? Compared to the previous question, what are the long-run implications for this economy?

Assume no population growth.